Predicting Loan Default Risk

Model Development & Evaluation

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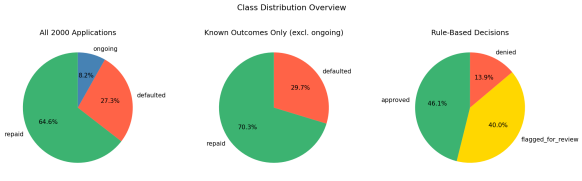
Exploratory Data Analysis

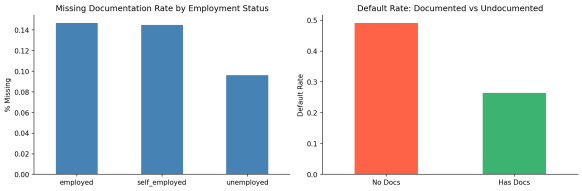
Target Imbalance & Data Scope

- Only about 15% of known outcomes are defaults. With that kind of imbalance, raw accuracy is misleading, so we use F1 and AUC instead.
- About 8% of applications are still `ongoing` with no final label, so they're excluded from training.
- The chart shows how small the default class is relative to repaid; any model that just predicts "repaid" would look 85% accurate.



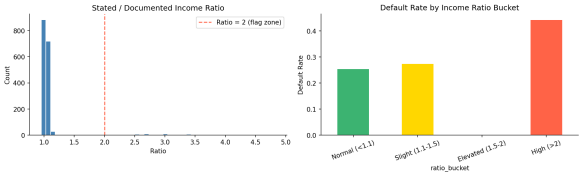
Missing Data as a Predictive Signal

- 15% of applications are missing `documented_monthly_income`.
- Missingness is not random; applicants without documentation default at a much higher rate.
- So missing docs isn't just a data quality problem; it's a genuine risk signal.



Behaviour Signals Over Raw Values

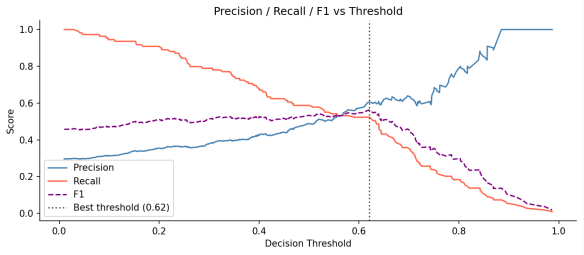
- When stated income is more than 2× documented income, default rates spike. Misrepresentation is a strong risk signal.
- Default rate jumps from ~17% to ~42% once the loan is more than 30% of monthly income.
- The raw dollar amount matters less than how the income *relates* to the loan size.



Model Choice

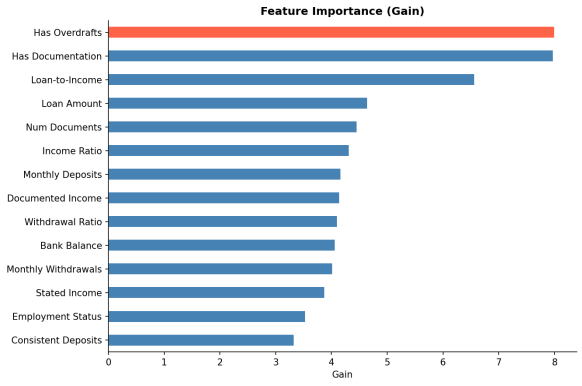
Why XGBoost?

- Picks up non-linear interactions between features — logistic regression can't do this.
- `scale_pos_weight` deals with class imbalance natively, no resampling needed.
- Threshold tuned for F1 on defaults, not the naive 0.5 cutoff.



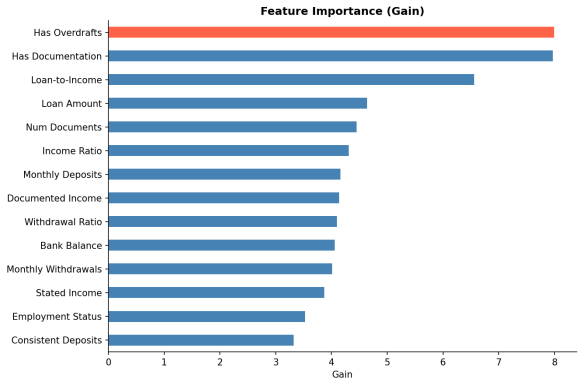
What Drives Default Risk?

- Gain-based importance pulled straight from the booster.
- `loan_to_income` and `withdrawal_ratio` top the list — ratios beat raw numbers, same story as EDA.
- `has_documentation` in top 5 — not having docs is a red flag on its own.



Explaining Individual Decisions

- Global importance $\neq$ why this applicant got flagged.
- Each denial surfaces the top features and actual values for that applicant — reviewers see the reasoning.
- Regulator asks why someone was denied? The explainer answers it automatically.



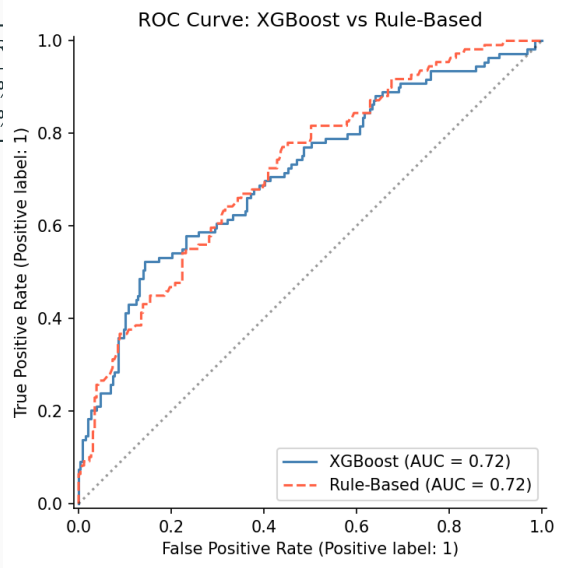
Evaluation

XGBoost vs. Rule-Based Baseline

	Rec.	F1	AUC	FPR	FNR
XGBoost	67.0%	53.5%	70.9%	35.1%	33.0%
Rule	33.0%	43.1%	72.2%	8.5%	67.0%

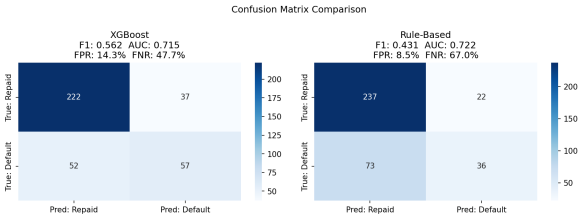
XGBoost cuts the missed-default rate (FNR) by half, from 67% to 33%.

The rule-based model looks more precise, but only because it barely flags anyone.



The Deploy-Tomorrow Trade-off

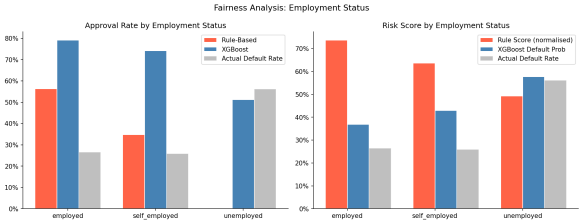
- XGBoost catches **+37** more defaults per 368 applications vs. the baseline (33% → 67% recall).
- Trade-off: **+69** additional good applicants wrongly denied (8.5% → 35.1% FPR).
- *Who decides?* The founders, weighing the cost of a missed default (full loan loss) against the cost of losing a customer (CAC). The threshold is a dial, not a fixed number.



Fairness & AI Ethics

Algorithmic Fairness & the Ethics of Automated Decisions

- The rule-based system penalises self-employed applicants even though their actual default rates are similar to employed applicants. That's textbook proxy discrimination.
- XGBoost learns from outcomes, not labels, so approval rates track real risk instead of employment category.
- Tension: keeping `employment_status` risks disparate-impact liability even if the model uses it correctly. Removing it may increase defaults.



Production

If This Went Live Tomorrow. . .

- **Model drift:** a recession doubling default rates would silently degrade predictions. The rule-based system at least degrades *transparently*.
- **Distribution monitoring:** feature distributions (e.g., `loan_to_income` under inflation) need to be tracked continuously. The static rule-based system doesn't need any of that.
- **Regulatory audits:** covered. The local explainer sits alongside the prediction API and generates human-readable denial reasons per applicant.